

Practice Test 2 (Challenge) · Answer Key

Chemical Bonding Unit Test · 80 points

How to use this key. This test is marked strictly on detail. In Part B, a drawing with one missing lone pair, bracket or charge loses that point, and an explanation without the underlined key terms loses the point for that idea.

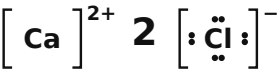
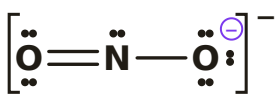
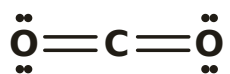
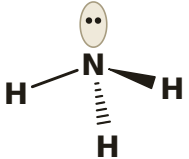
Part A · Multiple Choice (2 points each, 36 total)

Quick key: 1 C 2 B 3 D 4 A 5 C 6 B 7 D 8 C 9 D 10 A 11 B 12 C 13 D 14 B 15 C 16 D 17 A 18 C

#	Ans	Why
1	C	conducts electricity as a solid. Ionic solids conduct only when molten or dissolved. Watch the word NOT.
2	B	nonpolar covalent. $\Delta EN = 0$, so the pair is shared equally.
3	D	3. 2 bonded O atoms + 1 lone pair = 3. The double bond counts once (not 4), and the lone pair must be counted (not 2).
4	A	copper(I) bromide. Br is 1-, so the single copper ion is Cu^+ . Copper needs a numeral; ionic names take no prefixes.
5	C	8. $5 + 4 = 9$, then subtract 1 for the + charge. A positive charge means electrons were lost.
6	B	sulfite, SO_3^{2-}. NO_2^- is nitrite; carbonate is CO_3^{2-} ; ammonium is NH_4^+ .
7	D	tetrahedral. 4 domains on O (2 bonds + 2 lone pairs). Bent is the molecular geometry.
8	C	C-Cl. C-Cl: $3.0 - 2.5 = 0.5$. C-H is 0.4; N-Cl and P-H are 0.
9	D	$Fe_2(SO_4)_3$. Fe^{3+} and SO_4^{2-} : criss-cross to 2 and 3. The 3 belongs to sulfate, not to iron.
10	A	Its bonds are polar and the molecule is nonpolar. C-Cl has ΔEN 0.5 (polar), but the tetrahedral shape with four identical atoms makes the dipoles cancel.
11	B	1. 26 electrons: 3 P-Cl bonds, 3 lone pairs on each Cl, and 1 lone pair left for P. There are 10 lone pairs in the whole molecule.
12	C	covalent network. An ionic compound would conduct when molten. "In any state" rules it out.
13	D	NO_3^-. N in NO_3^- has 3 bonded atoms and no lone pair. The other three have a lone pair on the central atom, so they are trigonal pyramidal.
14	B	+1. $FC = 5 \text{ valence} - 0 \text{ lone-pair electrons} - 4 \text{ bonds} = +1$.
15	C	CH_2Cl_2. Its outer atoms are not all the same, so the C-Cl dipoles are not cancelled.
16	D	$MgCl_2$, magnesium dichloride. Ionic names never use prefixes: it is magnesium chloride.
17	A	8. 4 lone pairs (2 on each O) = 8 electrons. The question asks for electrons, not pairs.
18	C	X metallic, Y ionic, Z molecular covalent. Conducts in both states: metallic. Only when liquid: ionic. Never, with a low melting point: molecular covalent.

Part B · Free Response (44 points)

19. Find the mistakes (8 points: 1 for the mistakes, 1 for the correct drawing)

 A. correct	 B. correct (either resonance form)	 C. correct	 D. correct
--	--	---	--

A. Two mistakes: only **one** chloride ion is shown (Ca^{2+} needs two Cl^- to balance), and the chloride has **7 dots** (an ion that has gained an electron has 8).

B. Two mistakes: the **lone pair on N is missing** (the drawing shows 16 electrons; NO_2^- has $5 + 12 + 1 = 18$), and the **brackets and – charge** are missing.

C. One mistake: **C has only 4 electrons** (two single bonds). Move one lone pair from each O into a double bond: $\text{O}=\text{C}=\text{O}$, with 2 lone pairs on each O. The electron count (16) was already right.

D. Two mistakes: the **lone pair on N is missing**, and the **solid wedge is backwards** (the thin end goes at the central atom).

Award the "mistakes" point only if all the mistakes in that drawing are named.

20. Careful calculations (10 points)

(a) 1 point per row; the subtraction must be shown.

Bond	ΔEN	Type
N-H	$3.0 - 2.1 = 0.9$	polar covalent
C-Cl	$3.0 - 2.5 = 0.5$	polar covalent (0.5 is the start of the polar range)
P-H	$2.1 - 2.1 = 0$	nonpolar covalent
Mg-O	$3.5 - 1.2 = 2.3$	ionic

(b) 1 point per row (working and name both needed).

Formula	Working	Name
$\text{Fe}_2(\text{SO}_4)_3$	3 sulfate $\times 2- = 6-$, so 2 Fe = $6+$, each Fe^{3+}	iron(III) sulfate
Cu_2O	1 oxide = $2-$, so 2 Cu = $2+$, each Cu^+	copper(I) oxide
FeCO_3	1 carbonate = $2-$, so Fe^{2+}	iron(II) carbonate

(c) 1 point per row.

Ion	Working	Total
PO_4^{3-}	$5 + 4 \times 6 + 3$	32
H_3O^+	$3 \times 1 + 6 - 1$	8
HCO_3^-	$1 + 4 + 3 \times 6 + 1$	24

Common slips: adding for a + charge (H_3O^+ is not 10); using the subscript on the metal as its charge (Fe_2 does not mean Fe^{2+}); calling 0.5 nonpolar.

21. Molecular geometry and polarity (12 points)

Formula	Valence e^-	Lewis structure	Electron geometry	Molecular geometry & bond angle	3-D redraw	Bond polarity (show ΔEN)	Molecular polarity
HCN	10	$\text{H}-\text{C}\equiv\text{N}:$	linear (2 domains)	linear 180°		C-H: 0.4, nonpolar C≡N: 0.5, polar	Polar
SO_3^{2-}	26		tetrahedral (4 domains)	trigonal pyramidal $\approx 107^\circ$		S-O: 1.0, polar	Polar
COCl_2	24		trigonal planar (3 domains)	trigonal planar 120°		C=O: 1.0, polar C-Cl: 0.5, polar	Polar

4 points per row: 1 for valence electrons and Lewis structure; 1 for both geometries with the angle; 1 for the redraw; 1 for bond polarity with ΔEN and molecular polarity.

Details that are often missed. HCN: the triple bond counts as one domain, the lone pair is on N, and C-H is nonpolar while $\text{C}\equiv\text{N}$ is polar. SO_3^{2-} : add 2 electrons for the charge, keep the lone pair on S, and use brackets with 2^- ; the redraw needs the lone pair, 1 plain line, 1 wedge and 1 dash. COCl_2 : trigonal planar is flat, so plain lines only; it is polar because the outer atoms are not all the same.

22. Explain using the correct key terms (8 points)

(a) In the solid, the **ions** are in **fixed positions** in the lattice and cannot move. When it melts, the ions are **free to move** and carry the charge.

1 point each idea. "Electrons" instead of "ions" earns 0.

(b) In copper, the layers of positive ions **slide** past each other and the **delocalized electrons** still hold them together. In NaCl, a shifted layer brings **ions of the same charge** next to each other; they **repel** and the crystal splits.

1 point for copper, 1 for sodium chloride.

(c) CO_2 is **linear** (symmetric). The two $\text{C}=\text{O}$ **bond dipoles** are equal and point in opposite directions, so they **cancel**.

1 point: linear or symmetric shape. 1 point: dipoles cancel.

(d) Diamond is a **covalent network**: melting it means breaking strong **covalent bonds** throughout the solid. Dry ice is made of separate CO₂ molecules: only the **weak forces between molecules** are overcome, and the covalent bonds inside each molecule stay intact.
1 point for diamond, 1 for dry ice.

23. Designing materials (6 points)

(a) **Metallic** bonding. It **conducts electricity** because its **delocalized electrons** are free to move, and it is **ductile** (can be drawn into cable) because the ion layers slide without the bonding breaking. Also acceptable: low density, or the alloy is stronger than pure aluminum.
1 point: metallic. 1 point: two properties, each explained.

(b) In the solid the **ions** are **fixed in the lattice** and cannot move, and there are **no delocalized (free) electrons**, so there is nothing to carry charge through the disc.
1 point: ions cannot move. 1 point: no free electrons.

(c) It is **brittle**. A knock shifts a layer of ions so that **like charges** are next to each other; they **repel** and the disc cracks.
1 point: brittle. 1 point: explanation.